

Plan

- Basic Feasible Solution (BFS)
- Non degenerate BFS
- Degenerate BFS
- Simplex Algorithm

- Consider LPP(P)

Max or Min $\mathbf{c}^T \mathbf{x}$

subject to $\mathbf{A}_{m \times n} \mathbf{x} = \mathbf{b}_{m \times 1}, \mathbf{x} \geq \mathbf{0}$ (*)

where $\text{rank}(\mathbf{A}) = m$

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where $k > \text{rank}(\mathbf{A}) = m$

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- A **Basic Solution** may **not** be a **non negative** vector, so need not be a **feasible solution** of the LPP
- A **BFS** of a LPP of the form (*), can have **at most m** strictly positive components ($\text{rank}(A) = m$)

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- $B_{m \times m} = [\tilde{\mathbf{a}}_1 \dots \tilde{\mathbf{a}}_m]$ with columns $\tilde{\mathbf{a}}_i, i = 1, \dots, m$, of A is called **the basis matrix** corresponding to $\mathbf{x}$

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- The variables $x_1, \dots, x_m$ are called **basic variables**, and $x_{m+1} = x_{m+2} = \dots = x_n = 0$, are called **non basic variables** of the **BFS** $\mathbf{x}$

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- If LPP (**P**) is the **diet problem**, (with $\geq$ inequalities changed to equalities in the original problem),
then $\mathbf{x}$ gives the **quantities** of the food products F_j ,
 $j = 1, \dots, n$ in the diet

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- Then $\begin{bmatrix} u_{1k} \\ \vdots \\ u_{mk} \end{bmatrix} = B^{-1} \tilde{\mathbf{a}}_k$ for all $k = 1, 2, \dots, n$

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- For all $k = 1, \dots, m$, $z_k = c_k$
- If $z_k > c_k$ then look for a better solution

- the simplex table corresponding to BFS $\mathbf{x}$ is given by

	0	..	0	..	$c_s - z_s$	..	$c_k - z_k$	..	
	$B^{-1}\tilde{\mathbf{a}}_1$	..	$B^{-1}\tilde{\mathbf{a}}_m$	..	$B^{-1}\tilde{\mathbf{a}}_s$	..	$B^{-1}\tilde{\mathbf{a}}_k$	..	$B^{-1}\mathbf{b}$
$\tilde{\mathbf{a}}_1$	1	..	0	..	u_{1s}	..	u_{1k}	..	x_1
$\tilde{\mathbf{a}}_2$	0	..	0	..	u_{2s}	..	u_{2k}	..	x_2
$\vdots$	0	..	0	..	$\vdots$	..	$\vdots$	..	$\vdots$
$\tilde{\mathbf{a}}_r$	$\vdots$	..	$\vdots$	..	u_{rs}	..	u_{rk}	..	x_r
$\vdots$	$\vdots$	..	$\vdots$	..	$\vdots$	..	$\vdots$	..	$\vdots$
$\tilde{\mathbf{a}}_m$	0	..	1	..	u_{ms}	..	u_{mk}	..	x_m

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then $x'_i \geq 0$ only if $x'_s \geq 0$, is such that $x'_s \leq \frac{x_i}{u_{is}}$

- If $\mathbf{x}'$ is feasible for the LPP, then $x'_s \geq 0$ is such that $x'_i \geq 0$ for all $i = 1, \dots, m$
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- **Case 1a:** For atleast one $i = 1, \dots, m$, $u_{is} > 0$
(where s is as defined in **Case 1**)

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- $\mathbf{x}' \geq 0 \Rightarrow x'_s \leq \frac{x_r}{u_{rs}}$

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- $\mathbf{x}' \geq \mathbf{0} \Rightarrow \mathbf{x}'_s \leq \frac{x_r}{u_{rs}}$
- Also $A\mathbf{x}' = \mathbf{b}$ which implies $\mathbf{x}' \in \text{Fea}(LPP)$

- $\mathbf{c}^T \mathbf{x}' = \sum_{i=1}^m c_i x'_i + c_s x'_s$

- $$\mathbf{c}^T \mathbf{x}' = \sum_{i=1}^m c_i x'_i + c_s x'_s$$

$$= \mathbf{c}^T \mathbf{x} + x'_s (c_s - z_s)$$

- $$\begin{aligned}
 \bullet \quad \mathbf{c}^T \mathbf{x}' &= \sum_{i=1}^m c_i x'_i + c_s x'_s \\
 &= \mathbf{c}^T \mathbf{x} + x'_s (c_s - z_s) \\
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 \end{aligned}$$

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- $\mathbf{c}^T \mathbf{x}' < \mathbf{c}^T \mathbf{x}$ if $x'_s > 0$

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- $$\mathbf{c}^T \mathbf{x}' < \mathbf{c}^T \mathbf{x} \text{ if } x'_s > 0 \text{ which is when the minimum ratio } \frac{x_r}{u_{rs}} > 0$$

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- $\bullet x_s$ is called the **entering variable**,

- $\bullet \mathbf{c}^T \mathbf{x}' = \sum_{i=1}^m c_i x'_i + c_s x'_s$
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- $\bullet$ Let $x'_s = \frac{x_r}{u_{rs}}$
- $\bullet$ Then $x'_r = x_r - u_{rs} \frac{x_r}{u_{rs}} = 0$
- $\bullet x_s$ is called the **entering variable**, and x_r is called a **leaving variable**

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- $\bullet$ Let $x'_s = \frac{x_r}{u_{rs}}$
- $\bullet$ Then $x'_r = x_r - u_{rs} \frac{x_r}{u_{rs}} = 0$
- $\bullet x_s$ is called the **entering variable**, and x_r is called a **leaving variable**
- $\bullet$ If there exists $r, t \in \{m+1, \dots, n\}$
 $\frac{x_r}{u_{rs}} = \frac{x_t}{u_{ts}} = \min \left\{ \frac{x_i}{u_{is}} : u_{is} > 0 \right\}$
then take any **one** of r, t as the **leaving variable**

- x' is a **BFS** of the LPP

- $\mathbf{x}'$ is a **BFS** of the LPP
- The basis matrix corresponding to $\mathbf{x}'$ is
$$B' = [\tilde{\mathbf{a}}_1 \dots \tilde{\mathbf{a}}_{r-1} \tilde{\mathbf{a}}_s \tilde{\mathbf{a}}_{r+1} \dots \tilde{\mathbf{a}}_m]$$

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- $\mathbf{b} = \sum_{i=1, i \neq r}^m (x_i - \frac{u_{is}}{u_{rs}} x_r) \tilde{\mathbf{a}}_i + \tilde{\mathbf{a}}_s (\frac{x_r}{u_{rs}})$

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$$z'_k = z_k + \frac{(c_s - z_s)}{u_{rs}} u_{rk}$$

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- If z'_k denotes the new values of z_k then

$$z'_k = z_k + \frac{(c_s - z_s)}{u_{rs}} u_{rk}$$
and $c_k - z'_k = (c_k - z_k) - \frac{(c_s - z_s)}{u_{rs}} u_{rk}$

- The simplex table corresponding to the new BFS $\mathbf{x}'$ is given by

	$C_s - Z'_s$ $C_s - Z_s - \frac{(C_s - Z_s)}{u_{rs}} u_{rs} = 0$	$C_k - Z'_k$ $(C_k - Z_k) - \frac{(C_s - Z_s)}{u_{rs}} u_{rk}$	
	$B'^{-1} \tilde{\mathbf{a}}_s$	$B'^{-1} \tilde{\mathbf{a}}_k$	$B'^{-1} \mathbf{b}$
$\tilde{\mathbf{a}}_1$	$u_{1s} - \frac{u_{1s}}{u_{rs}} u_{rs} = 0$	$u_{1k} - \frac{u_{1s}}{u_{rs}} u_{rk}$	$x_1 - \frac{u_{1s}}{u_{rs}} x_r$
$\tilde{\mathbf{a}}_2$	$u_{2s} - \frac{u_{2s}}{u_{rs}} u_{rs} = 0$	$u_{2k} - \frac{u_{2s}}{u_{rs}} u_{rk}$	$x_2 - \frac{u_{2s}}{u_{rs}} x_r$
$\vdots$	0	$\vdots$	$\vdots$
$\tilde{\mathbf{a}}_s$	$\frac{u_{rs}}{u_{rs}} = 1$	$\frac{u_{rk}}{u_{rs}}$	$\frac{x_r}{u_{rs}}$
$\vdots$	$\vdots$	$\vdots$	$\vdots$
$\tilde{\mathbf{a}}_m$	$u_{ms} - \frac{u_{ms}}{u_{rs}} u_{rs} = 0$	$u_{mk} - \frac{u_{ms}}{u_{rs}} u_{rk}$	$x_m - \frac{u_{ms}}{u_{rs}} x_r$

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	$B'^{-1} \tilde{\mathbf{a}}_s$	$B'^{-1} \tilde{\mathbf{a}}_k$	$B'^{-1} \mathbf{b}$
$\tilde{\mathbf{a}}_1$	$u_{1s} - \frac{u_{1s}}{u_{rs}} u_{rs} = 0$	$u_{1k} - \frac{u_{1s}}{u_{rs}} u_{rk}$	$x_1 - \frac{u_{1s}}{u_{rs}} x_r$
$\tilde{\mathbf{a}}_2$	$u_{2s} - \frac{u_{2s}}{u_{rs}} u_{rs} = 0$	$u_{2k} - \frac{u_{2s}}{u_{rs}} u_{rk}$	$x_2 - \frac{u_{2s}}{u_{rs}} x_r$
$\vdots$	0	$\vdots$	$\vdots$
$\tilde{\mathbf{a}}_s$	$\frac{u_{rs}}{u_{rs}} = 1$	$\frac{u_{rk}}{u_{rs}}$	$\frac{x_r}{u_{rs}}$
$\vdots$	$\vdots$	$\vdots$	$\vdots$
$\tilde{\mathbf{a}}_m$	$u_{ms} - \frac{u_{ms}}{u_{rs}} u_{rs} = 0$	$u_{mk} - \frac{u_{ms}}{u_{rs}} u_{rk}$	$x_m - \frac{u_{ms}}{u_{rs}} x_r$

- The entry u_{rs} of the previous table which is made 1 (by dividing) in this table is called the **pivot element**

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 $\Rightarrow \mathbf{x}' \in \text{Fea}(LPP)$ for all $x'_s \geq 0$

- **Case 1b:** For all $i = 1, \dots, m$, $u_{is} \leq 0$ (where s is as defined in **Case 1**)
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 So given any $M \in \mathbb{R}$, by taking $x'_s > 0$ sufficiently large we can make $\mathbf{c}^T \mathbf{x}'$ smaller than M

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 So given any $M \in \mathbb{R}$, by taking $x'_s > 0$ sufficiently large we can make $\mathbf{c}^T \mathbf{x}'$ smaller than M
 So the LPP has **NO optimal solution**

- $D = \{\mathbf{d} \in \mathbb{R}^n : \mathbf{d} \neq \mathbf{0}, A_{m \times n} \mathbf{d} = \mathbf{0}, \mathbf{d} \geq \mathbf{0}\}$ is the set of all directions of $S = \text{Fea}(LPP) = \{\mathbf{x} \in \mathbb{R}^n : A_{m \times n} \mathbf{x} = \mathbf{b}, \mathbf{x} \geq \mathbf{0}, \text{rank}(A) = m\}$

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- If for some basis matrix B and a column $\tilde{\mathbf{a}}_s$ of A , $B^{-1} \tilde{\mathbf{a}}_s \leq \mathbf{0}$

$$\mathbf{d} = \begin{bmatrix} -B^{-1} \tilde{\mathbf{a}}_s \\ 0 \\ \vdots \\ 0 \\ 1 \\ 0 \\ \vdots \\ 0 \end{bmatrix}$$

is an extreme direction of S ,

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is an extreme direction of S , where the entry **1** in the above vector is at the s th position

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is an extreme direction of S , where the entry **1** in the above vector is at the s th position

- If $\mathbf{d}$ is as above then $\mathbf{c}^T \mathbf{d} < 0$

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$$\frac{x_r}{u_{rs}} = \min\left\{\frac{x_i}{u_{is}} : u_{is} > 0\right\}$$

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- The above **optimality conditions** (for max and min problems) are **sufficient** but not **necessary** for an optimal solution

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Min $x_1 + x_2$ subject to

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The simplex table corresponding to the **BFS** with basic variables x_2 and s_1 is given by:

$c_j - z_j$	2	0	0	-1	
	x_1	x_2	s_1	s_2	$B^{-1}\mathbf{b}$
x_2	-1	1	0	1	1
s_1	1	0	1	-2	2

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$c_j - z_j$	1	1	0	0	
	x_1	x_2	s_1	s_2	$B^{-1}\mathbf{b}$
s_2	-1	1	0	1	1
s_1	-1	2	1	0	4

Note that the above table is optimal

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|-------------|-------|-------|-------|-------|--------------------|
| | x_1 | x_2 | s_1 | s_2 | $B^{-1}\mathbf{b}$ |
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|-------------|-------|-------|-------|-------|--------------------|
| | x_1 | x_2 | s_1 | s_2 | $B^{-1}\mathbf{b}$ |
| x_2 | 0 | 1 | 1 | -1 | 3 |
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The above table is optimal

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x_2	-1	1	0	1	0	1
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	x_1	x_2	s_1	s_2	s_2	$B^{-1}\mathbf{b}$
s_2			0	1	0	4
s_1			1	0	0	1
s_3			0	0	1	9

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